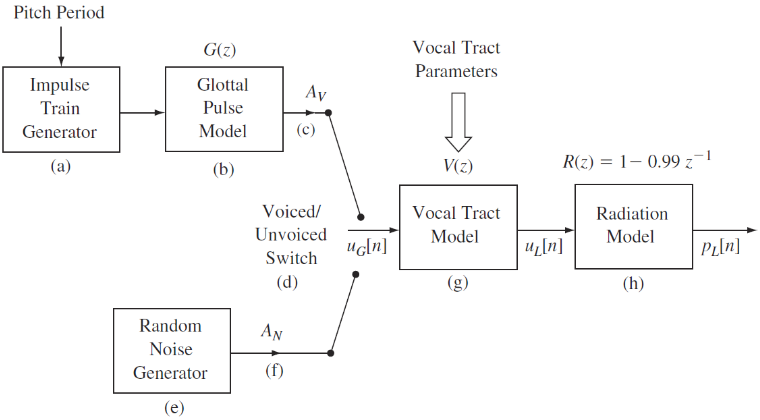
Source-Filter Model of Speech Production

Overview

- Speech production modeled as a **source-filter system**
- Source = excitation (voiced/unvoiced)
- Filter = vocal tract shaping
- Radiation = lip output effect



Impulse Train Generator

Equation:

$$s[n] = \sum_k \delta[n - kT]$$

Generates periodic impulses

T = pitch period

Models vocal fold vibration (voiced sounds)

Glottal Pulse Model

Equation:

$$U_G(z) = G(z)S(z)$$

Shapes impulses into realistic waveform

Introduces spectral tilt

Models airflow through vocal folds

Voiced / Unvoiced Excitation

Voiced: periodic signal

Unvoiced: noise-like signal

Noise model:

$$u_n[n] \sim \mathcal{N}(0, \sigma^2)$$

Switch selects appropriate excitation

Gain Control

Voiced gain:

$$u_v[n] = A_v u_G[n]$$

Noise gain:

$$u_n[n] = A_N w[n]$$

Controls signal energy

Vocal Tract Model

Transfer function:

$$V(z) = \frac{1}{1 + \sum_{k=1}^{p} a_k z^{-k}}$$

- All-pole filter
- Models resonances (formants)
- Shape depends on articulators

Radiation Model

Equation:

$$R(z) = 1 - 0.99z^{-1}$$

High-pass effect

Models lip radiation

Overall System

Complete model:

$$P_L(z) = R(z)V(z)G(z)S(z)$$

Source-filter cascade

Linear time-invariant approximation

Formants and Poles

Formants = spectral peaks

Correspond to poles of $V(z)$

Poles near unit circle $\Rightarrow$ sharp resonance

Pole representation:

$$z_k = r_k e^{j\omega_k}$$

ω_k = formant frequency

r_k = bandwidth control

Physical Interpretation

Changing vocal tract shape moves poles

Different vowels = different pole locations

Example:

- /a/: low F1, mid F2
- /i/: low F1, high F2

Connection to LPC

Linear Predictive Coding (LPC) estimates a_k

Provides pole locations

Used in:

- Speech coding
- Speech recognition

Conclusion

Speech = excitation + filtering

Formants are key perceptual features

Source-filter model widely used in DSP